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Physics-informed machine learning prediction of char mass evolution in the catalytic pyrolysis of polyetherimide/graphite nanocomposites

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ABSTRACT

Carbonaceous structures can be produced via pyrolysis of polymeric precursors for applications in gas separation membranes, energy storage, flexible electronics, electromagnetic interference shielding foams, etc. Maximizing char yield is a primary objective, determined by precursor chemistry, composition, pyrolysis conditions, and kinetics. The complex, non-linear relationships among these factors favor machine learning (ML) for process design and optimization. A physics-informed, transformer-based ML model was developed to predict char mass evolution (thermal decomposition) of transition-metal-catalyzed polyetherimide (PEI)/graphite (Gr) nanocomposites from thermogravimetric analysis (TGA) data. The dataset included 38 formulations with varying Gr and catalyst (Fe, Ni, Co) contents, heating rates, and pyrolysis temperatures. Additional features captured Gr and catalyst structural and electronic properties (crystal system, d-orbital free electrons, lattice parameters, cohesive energy, carbide formation energy, electrical conductivity at 20 °C) and kinetic parameters from 2D/3D Avrami–Erofeev models (pre-exponential factor, activation energy). Data were split into "seen" catalysts (Fe, Ni) for training/validation and an "unseen" catalyst (Co) for testing. Hyperparameters and feature selection were optimized via the random forest method. The model achieved $R^2 > 0.98$ on unseen data, accurately predicting TGA curves and kinetic trends. Experimental and ML-predicted curves showed close agreement, with successful extrapolation to Co-containing nanocomposites. This study integrates kinetics modeling with advanced ML to enhance prediction of pyrolysis behavior in polymer nanocomposites, providing a practical framework for developing carbonaceous materials with tailored properties.

1. Introduction

Polyetherimide (PEI) [1], typically branded as Ultem©, is a thermoplastic polymer with a robust molecular structure and an aromatic-rich backbone (chemical formula of the repeating unit: $C_{37}H_{24}O_6N_2$; MW: 592.61 g/mol; glass transition temperature: 217 °C) that makes it an ideal polymer precursor for producing carbonaceous materials for a variety of applications, including gas separation membranes [2–6], energy storage (supercapacitors and battery electrodes) [7], flexible electronics [8], electromagnetic interference shielding foams [9], etc. During pyrolysis, PEI undergoes a series of thermal decomposition processes that lead to the formation of high carbon content residue (char) with good structural integrity [10]. This outcome is attributed to reduced volatile loss during PEI pyrolysis because of its high aromatic content [11]. Thermogravimetric analysis (TGA) [12] and kinetic modeling studies [13] reveal that PEI and other similar polymers undergo thermal decomposition via a combination of random chain scission, formation of stable carbon networks by aromatic fragments, and volatile removal. In general, TGA is commonly used to evaluate the thermal stability of polymers and composites and to analyze their thermal decomposition kinetics [14].

Efforts have been expended on increasing the char yield of polyimide

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(PI) and PEI, as well as improving the properties of carbonized polymers by incorporating nanofillers, such as graphene or graphite [15,16], carbon nanotubes [17], metal oxide nanoparticles, such as titanium dioxide (TiO_2) [18], and transition metal catalysts, such as Fe, Co, or Ni [19]. Overall, nanofillers may improve the thermal stability of PEI, create a barrier effect on the release of volatile compounds during pyrolysis, reduce the decomposition rate, and promote the formation of a thermally stable char layer on the polymer surface [20]. The latter acts as a protective barrier, reducing heat and mass transfer during pyrolysis, thereby enhancing flame retardancy and reducing mass loss. Some nanofillers may act as catalyst during pyrolysis, altering the PEI decomposition pathways. The incorporation of nanofillers can also improve the mechanical integrity of the char formed during pyrolysis, which is crucial in applications requiring structural stability at high temperatures such as carbonaceous membranes.

The methods for determining the kinetic parameters from thermal analysis data generally fall into two categories: (1) "model-free" and (2) model fitting approaches. The model-free approach does not assume a specific reaction model and focuses on activation energy as a function of conversion. On the other hand, the model fitting approach assumes a fixed reaction mechanism with constant activation energy throughout the process. However, a limitation of this method is its reduced accuracy when dealing with complex reactions, which involves fitting conversion-time data or conversion-rate data to specific models to determine reaction orders, rate constants, and activation energy [21]. While these kinetic models are useful tools to predict the pyrolysis kinetics of polymers or polymer nanocomposites (PNCs), a comprehensive modeling of the precursor formulation, pyrolysis conditions, catalysis performance (if catalyst is present in the precursor formulation), and other influencing factors on the char formation favors the use of a machine learning (ML) approach that can efficiently navigate the relationships between these factors.

With respect to the prediction of char yields and properties from various feedstocks such as biomass, ML has become an invaluable tool [22–28]. For example, by leveraging data on biomass characteristics and processing conditions, ML models can efficiently forecast pyrolysis outcomes, reducing the need for extensive experimental trials. Several ML algorithms have been employed to predict biomass char yields and their properties [29–31]: (1) random forest (RF), which is widely used for its robustness and accuracy and can achieve high R^2 values (up to 0.98) in predicting hydrochar yields and other properties, (2) gradient boosting machines (GBM), including variants such as XGBoost and AdaBoost, which have shown strong performance in modeling biochar yields with R^2 values ranging from 0.84 to 0.94, (3) artificial neural networks (ANNs), in particular multi-layer perceptron neural networks (MLP-NNs), which have demonstrated impressive accuracy, achieving R^2 values up to 0.96 in biochar yield prediction, and (4) hybrid models, which combine ML algorithms with optimization techniques such as particle swarm optimization (PSO) with enhanced predictive performance. For example, ensembled learning tree (ELT) integrated with PSO has achieved an R^2 value of 0.99 in biochar yield prediction. These ML models can capture complex patterns and relationships in the data, providing more flexible and accurate modeling for various applications compared to traditional approaches like model fitting [32]. In general, classification techniques can be employed to select the best type of catalyst, while regression methods are useful for predicting activation energy, material characteristics, and its physicochemical properties. They are particularly effective in predicting TGA and differential thermogravimetric (DTG) curves at varying heating rates during combustion processes of nanocomposites, biomass residues, and waste materials [33–37].

Based on the published ML models, several critical factors have been identified to influence char formation and its properties for the pyrolysis of a biomass feedstock: (1) pyrolysis temperature, which has consistently been recognized as the most significant variable affecting char yield and its quality, (2) residence time, which can influence the degree of carbonization and, consequently, the yield and properties of the resulting char [38], (3) biomass composition (proportions of lignin, cellulose, and hemicellulose in the biomass feedstock), where structural information has been found to be more predictive than elemental composition for biochar yield [31], and (4) ash and nitrogen content, where higher ash content generally negatively affects char yield, while nitrogen content can influence nitrogen retention in the char [39]. To enhance the transparency of ML models, interpretability techniques like Shapley additive explanations (SHAP) [40] have been employed. SHAP analysis helps in understanding the contribution of each input feature to the model predictions, thereby providing insights into the underlying processes influencing char production [41].

While some ML models have been developed for the pyrolysis of polymers [42], to the best of our knowledge, there is a lack of predictive models for the catalytic pyrolysis of PNCs, as well as a framework to understand the relative importance of the influence of different factors on the char yield. To address this scientific gap, a dataset of TGA data (400–1000 °C), extracted from the thermograms of 38 differently formulated PEI-based nanocomposites, containing graphite (Gr) (0–100 parts per hundred resin (phr), based on PEI), and transition metal catalysts (Fe, Co, and Ni), obtained at different heating rates (10–20 °C/min) under nitrogen (inert atmosphere), plus the parameters of two kinetic models, i.e., 2D and 3D Avrami-Erofeev equations [43], was used herein to develop a physics-informed transformer-based neural network model to predict the thermal decomposition behavior for various PEI or PNC formulations. The goal was to estimate "char mass" (solid mass at a given temperature) as a function of temperature (char mass evolution) by leveraging both sequential data, static catalyst properties, and physics-informed kinetic model features. The above model was selected due to the ability of its attention mechanism to capture long-range dependencies in time-series data without the vanishing gradient problem inherent to recurrent architectures such as long short-term memory (LSTM) networks or gated recurrent unit (GRU) networks. The model designed for the task of predicting the TGA data is simplified from the traditional transformer-based neural network model [44], which was developed for natural language processing. Instead, this model excludes decoders and utilizes a fully connected layer to project to the output dimension for char mass regression. The novel contribution of this architecture lies in the combination of static and dynamic input features. Temperature inputs are modulated by a cross-attention mechanism to incorporate the static features as learned contextual information at each point during the TGA curve prediction process, which allows the model to dynamically utilize the features that are most important during each stage of pyrolysis using the existing architectural components of the transformer. Additionally, the model incorporates physics-informed loss terms, which penalize non-monotonic model outputs because of mass conservation. Overall, the transformer-based architecture presents a novel and effective approach for predicting TGA curves that generalize across different catalysts. The ML-guided optimization in this work enables the development of next-generation carbonaceous materials for advanced technological applications.

2. Methodology

2.1. Data collection and feature selection

The TGA data was obtained in our previous work [19], which involved a comprehensive investigation of the effects of the PEI/-graphite/catalyst formulation and TGA pyrolysis conditions on the resulting char yields within the framework of response surface methodology [45]. The original data [19] (Table 1) were supplemented herein with additional TGA data, Gr and catalyst-related data (Table 2), as well as 2D and 3D Avrami-Erofeev kinetic model [43] parameters for the sample decomposition rate, which were calculated using the Kinetics Neo v2.6.8 software (NETZSCH Group). Hereafter, reference is made to

Table 1
Summary of the original data[19] used for the development of the ML model.

type of data	range	actual values for collected data
precursor formulation:		
Gr content	0–100 phr[a]	0, 33, 50, 100 phr
catalyst type[b]	Fe, Co, Ni	-
catalyst content	-	0, 10 phr
TGA pyrolysis condition:		
heating rate	10–20 °C/min	10, 13, 15, 17, 20 °C/min
Temperature[b]	400–1000 °C	all char mass values from the thermograms

[a] parts per hundred resin (in this case, PEI).
[b] The initial mass loss due to the volatilization of the samples were ignored and mass percentages were shifted to 100 % at 400 °C.

Table 2
Additional graphite and catalyst related data used for the development of the ML model.

property	Gr	Fe	Co	Ni
crystal system[a]	H	BCC	FCC	FCC
d-orbital free electrons[b]	-	5	3	2
lattice parameter[a] (a and b; $a \cong b$) (nm)	0.25	0.29	0.35	0.35
cohesive energy[c] (eV)	7.37	4.28	4.39	4.44
carbide formation energy[d] (eV/atom)	-	0.060	0.109	0.084
electrical conductivity at 20 °C[e] (× 10^7 S/m)	0.03	1.00	1.79	1.43

[a] H: hexagonal; BCC: body-centered cubic (metallic α-Fe below 912 °C); FCC: face-centered cubic (metallic Co above 417 °C and metallic Ni below 1455 °C).
[b] Fe, Co, and Ni refer to $Fe(NO_3)_3{\cdot}9H_2O$ (Fe^{3+}), $Co(NO_3)_2{\cdot}6H_2O$ (Co^{2+}), and $Ni(NO_3)_2{\cdot}6H_2O$ (Ni^{2+}), respectively.
[c] Metallic α-Fe, Co, and Ni from Ref [46].
[d] Data retrieved from the Materials Project (https://materialsproject.org), material IDs: mp-1871 for Fe_2C, mp-22,488 for Co_2C, and mp-7586 for Ni_3C (accessed July 2025).
[e] In-plane value for Gr ($\rho = 2.3$ g/cm^3) from Ref [47]; Values for Fe, Co, and Ni derived from Ref [48].

the samples based on their designated codes, identifying their Gr content, pyrolysis heating rate, and type of catalyst used in their formulation (if any). For example, the sample containing 50 phr Gr and Co catalyst, heated at a rate of 10 °C/min was coded as G50-H10-Co. In total, 108,973 data points were used in the development of the ML model.

2.2. Kinetic modeling of the sample decomposition rate

The general rate equation used for the solid-state kinetics (in this case, PNC decomposition) is given as [43]:

$$\frac{d\alpha}{dt} = f(\alpha)k(T) \tag{1}$$

where α is the degree of conversion (mass-based), $k(T)$ is the temperature-dependent reaction rate constant, and $f(\alpha)$ is the reaction model (mechanism function). The temperature dependence of $k(T)$ is described by the Arrhenius equation:

$$k(T) = Ae^{\left(-\frac{E_a}{RT}\right)} \tag{2}$$

where A is the pre-exponential factor, E_a is the activation energy, R is the gas constant, and T is the absolute temperature. The general Avrami-Erofeev model (A_n) assumes random nucleation and growth of reaction sites with dimensionality n describing the growth mode:

$$f(\alpha) = n(1-\alpha)[-\ln(1-\alpha)]^{\frac{n-1}{n}}. \tag{3}$$

For polymer pyrolysis, 2D (A_2) or 3D (A_3) is common [49–51], depending on the mechanism (random scission vs end-chain scission), formation of intermediate char or gases, and catalyst presence. These models describe sigmoidal reaction behavior, where the initial stage (low α) signifies nucleation domination over reaction, the mid-stage (fast reaction) denotes growth acceleration as nuclei increase, and the final stage (high α) signifies slowing down of reaction as material is consumed. Herein, both models were used to predict the PNC decomposition rate. For this purpose, $\alpha(T)$ was calculated based on the sample thermograms and A_2 and A_3 models were fit to the data. All fit models had R^2 values of 0.98 and higher. After model fitting, the kinetic model parameters, i.e., pre-exponential factor (A) and activation energy (E_a), were extracted (**Tables S1** and **S2** in Supporting Information) and used as additional data in the development of the ML model. The A_2 and A_3 model fitting results for Sample G0-H17-Co are provided as examples in Fig. 1. Table 3 summarizes the selected features used in this study.

2.3. Data preprocessing

Prior to model training, the collected data underwent a comprehensive preprocessing pipeline to ensure data quality and consistency. For feature preparation, categorical variables representing the crystal system (CS1 and CS2) were encoded using ordinal encoding, which maps categorical values to integer representations. All numeric features were normalized using standardization (z-score normalization), which transforms each feature to have zero mean and unit variance. The d-orbital free electrons (dFE) were not normalized, as they physically represent integer values. This standardization ensures that features with different units and scales contribute without bias to the model training process.

2.4. Feature correlation analysis

Input features with high linear correlation (magnitude greater than 0.9) were filtered to minimize collinearity, reducing data requirements and ensuring that the model captures independent contributions from each variable. The linear correlation threshold of 0.9 was selected based on established practices in multicollinearity detection and feature selection for ML applications [52]. If two features exhibit a correlation over 0.9, only one is kept. Correlation between two variables, X and Y, is quantified using Pearson's correlation coefficient, given by:

$$r = \frac{\sum (X_i - \overline{X})(Y_i - \overline{Y})}{\sqrt{\sum (X_i - \overline{X})^2 \sum (Y_i - \overline{Y})^2}} \tag{4}$$

where r ranges from −1 to 1 with values close to 1 indicating a strong positive correlation and values close to −1 indicating a strong negative correlation.

2.5. Random forest for feature importance

In this work, the random forest (RF) algorithm was used to calculate feature importances. RF is an ensemble ML technique that builds multiple decision trees and combines their outputs to improve predictive performance and robustness. Each decision tree consists of a series of if-then decisions that minimize prediction loss, in this case, mean squared error (MSE), on the dataset. Randomness is introduced by training trees on a different bootstrap sample of the data and by randomly selecting subsets of the features at each split. Feature importance is calculated by measuring the average decrease in node impurity (e.g., entropy or Gini impurity) contributed by each feature, indicating how influential a feature is in improving model accuracy. The RF model was trained to predict mass loss as a function of temperature and the static input features, and the feature importances were calculated and visualized to identify the most important features for prediction of mass loss [53,54].

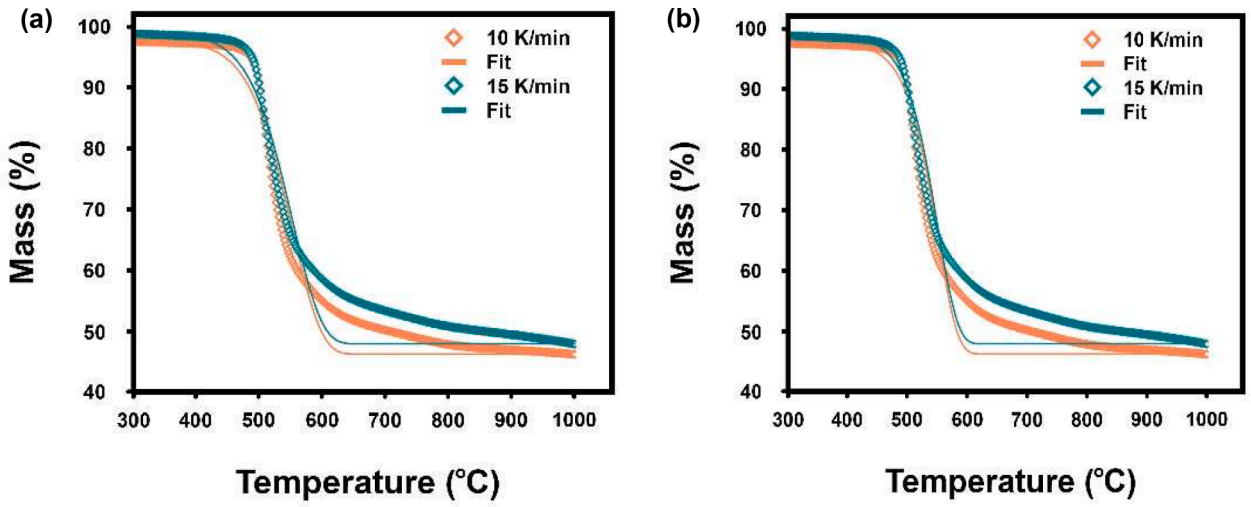


Fig. 1. Fitting results for (a) 2D (A_2) and (b) 3D (A_3) Avrami-Erofeev models for Sample G0-H17-Co (average $R^2 = 0.9812$).

Table 3
Features for the ML model development.

feature	acronym
formulation related:	
Gr content	GC
catalyst content	CC
pyrolysis condition related:	
heating rate	HR
temperature	T
catalyst (1) and Gr (2) related:	
crystal system	CS1
	CS2
d-orbital free electrons	dFE
lattice parameter	LP1
	LP2
cohesive energy	CE1
	CE2
carbide formation energy	CFE
electrical conductivity	EC
kinetic model related:	
pre-exponential factor	A2D
	A3D
activation energy	E_a2D
	E_a3D

2.6. Physics-informed transformers for char mass evolution prediction

The implemented architecture is simplified from the original transformer architecture. The primary modifications include the introduction of several physics-informed components, which include the physics-informed kinetic modeling features derived from the Avrami-Erofeev equations and a custom physics-informed loss function that enforces conservation of mass. A standard sinusoidal positional encoding was utilized during the encoding process. The input sequence $X_{dyn} \in \mathbb{R}^{T\times d}$ (where T is the sequence length and d is the feature dimension) is first projected into a higher-dimensional space:

$$h_0 = W_{dyn}X_{dyn} + P \tag{5}$$

where W_{dyn} is a learnable projection matrix and P is the positional encoder using *sine* and *cosine* functions. This ensures that the model retains the temporal order of the inputs, which is crucial to time-series modeling:

$$P_{(t,2i)} = \sin\left(\frac{t}{10000^{2i/d}}\right),\ P_{(t,2i+1)} = \cos\left(\frac{t}{10000^{2i/d}}\right) \tag{6}$$

The transformer encoder employs multi-head self-attention to capture dependencies across time steps. Given embeddings h, the self-attention mechanism computes:

$$\text{Attention}(Q,K,V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{7}$$

where Q, K, and V are the queries, keys, and values, respectively, computed by:

$$Q = W_Q h,\ K = W_K h,\ V = W_V h \tag{8}$$

where W_Q, W_K, and W_V are learnable weight matrices, and d_k is the feature dimension of the keys and queries. Multi-head attention enhances model expression by utilizing n attention layers in parallel, concatenating the outputs, and projecting the output with a learnable matrix W_O:

$$\text{MultiHeadAttention}(Q,K,V) = \text{Concat}(\mathit{Attention}_1,\ldots,\mathit{Attention}_n)W_O \tag{9}$$

Unlike the standard transformer, this model incorporates static catalyst properties via a cross-attention mechanism. The static input X_{static} is projected as:

$$h_{static} = W_{static}X_{static} \tag{10}$$

which is used as both the key and value in each attention layer, allowing the model to modulate the sequence representation based on static catalyst properties:

$$\text{CrossAttention}(h,h_{static}) = \text{softmax}\left(\frac{hW_Q(h_{static}W_K)^T}{\sqrt{d_k}}\right)(h_{static}W_V) \tag{11}$$

Rather than undergoing a decoding step typical of transformers utilized in natural language processing, which is unnecessary for low-dimensional regression tasks, the output of the attention layers is normalized via layer normalization and passed through a fully connected layer:

$$y_{pred} = W_{out}h_{final} + b \tag{12}$$

where W_{out} and b are learnable parameters. For the transformer model, the hidden dimension was 64, with two transformer layers, four attention heads, and a feedforward network size of 16. A dropout rate of 0.1 was applied to improve generalization. In Fig. 2, a schematic diagram of the proposed ML architecture is shown. The total number of trainable parameters was 37,009.

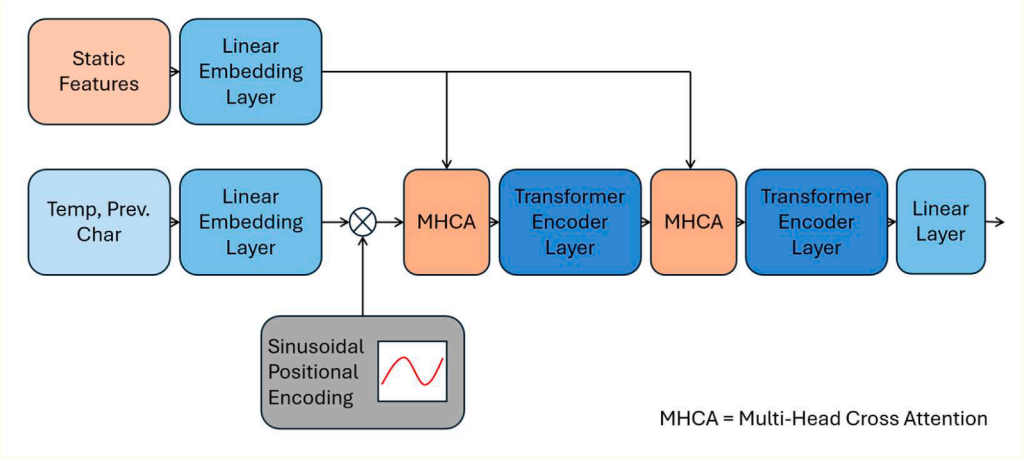


Fig. 2. Schematic diagram of the proposed neural network architecture (37,009 parameters).

2.7. Training and evaluation of the model

Catalyst parameters with high correlation (magnitude greater than 0.9) were filtered to minimize collinearity, ensuring that the model captures independent contributions from each variable. To ensure rigorous evaluation and prevent data leakage, an experiment-wise data splitting strategy was employed where each complete pyrolysis experiment (comprising the full temperature-time-conversion curve) was treated as an indivisible unit. Specifically, each of the experimental runs conducted across the different catalyst systems was assigned in its entirety to either the training, validation, or test set. The dataset was split as follows: 90 % for training, 10 % for validation, and the Co experiments were held out to show generalization on an "unseen" catalyst. Importantly, all data points from a given experiment, regardless of time or temperature, remained together in the same split to prevent the model from exploiting temporal correlations or experimental artifacts that could artificially inflate performance metrics. This approach is more stringent than random point-wise splitting, as it requires the model to generalize across entirely different experimental conditions rather than interpolating within partially observed experiments. This splitting strategy directly addresses concerns about data leakage and ensures that reported performance metrics reflect true generalization capability to unseen experimental conditions.

The model was trained using the AdamW optimizer [55] with a scheduled learning rate. Instead of using only mean absolute error (*MAE*), two custom physics-informed penalty terms were introduced based on the characteristics of TGA curves. Both physics-informed loss terms enforce conservation of mass on the predicted TGA curves. First, a monotonicity penalty (L_{mono}) is calculated:

$$L_{mono} = \lambda_{mono} \sum_{t=1}^{T-1} \max\left(0, y_{pred,t} - y_{pred,t-1}\right) \tag{14}$$

where λ_{mono} is a hyperparameter. Since char yield should be non-increasing with temperature, instances where predictions violate this constraint were penalized. The next penalty introduced was an over-prediction penalty L_{over}:

$$L_{over} = \lambda_{over} \sum_{t=1}^{T} \max\left(0, y_{pred,t} - 100\right) \tag{15}$$

where λ_{over} is a hyperparameter. Since char yield cannot exceed the initial value of 100 %, instances where char yield values exceed the initial value were additionally penalized. The total loss function was given by:

$$L = MAE\left(y_{pred}, y_{true}\right) + L_{mono} + L_{over} \tag{16}$$

The monotonicity penalty $\lambda_{mono} = 100$ was set to ensure that char yield predictions did not increase unexpectedly with temperature, and the overprediction penalty $\lambda_{over} = 1000$ penalized predictions exceeding 100 % char yield. The learning rate was set at 0.001496, with an AdamW optimizer incorporating a weight decay of 2×10^{-7} to mitigate overfitting. The learning rate scheduler adjusted the learning rate dynamically, reducing it by a factor of 0.8 if validation loss did not improve for 10 epochs. Early stopping was applied with a patience of 50 epochs to prevent excessive training. The transformer-based neural network hyperparameters, including number of layers, layer sizes, and number of attention heads, are summarized in **Table S3** in Supporting Information. All hyperparameters, including architectural design and physics-based regularization penalties, were determined through an iterative grid-search hyperparameter optimization process, where the initial optimization searches coarse values, and finer parameter choices were searched in later iterations.

3. Results and discussion

3.1. Feature correlation and importance

Linear correlation between the features (Table 3) was calculated and the resulting feature correlation heatmap is displayed in Fig. 3. Overall, several features with high positive correlation, including certain catalyst-related features, were observed; however, no features exhibited a strong negative correlation. For example, electrical conductivity (EC) exhibited a high correlation of 0.93 with the lattice parameter of Gr (LP1). Also, activation energy (E_a) exhibited a correlation of 1.00 with pre-exponential factor (A), which is consistent with their expected interdependency. Features with correlations above 0.90 with other features were considered duplicates and removed for the purposes of later ML steps. The features remaining after this correlation-based filtering step were T, GC, HR, CS1, CS2, LP2, A3D, CE1, dFE, EC, and CFE.

In Fig. 4, the RF correlations and importance of the features are highlighted. The influence of temperature (T), the primary independent variable for the TGA data, was excluded for the purposes of visualization. The most influential feature, as expected, is the Gr content (GC), which significantly outweighs other features in importance (Fig. 4**a**). By excluding GC, the influences of the other features become apparent (Fig. 4**b**). The high importance of GC aligns the critical role of Gr in catalytic pyrolysis through its well-established barrier effect [56], which directly influences the thermal and mass transfer environment of the

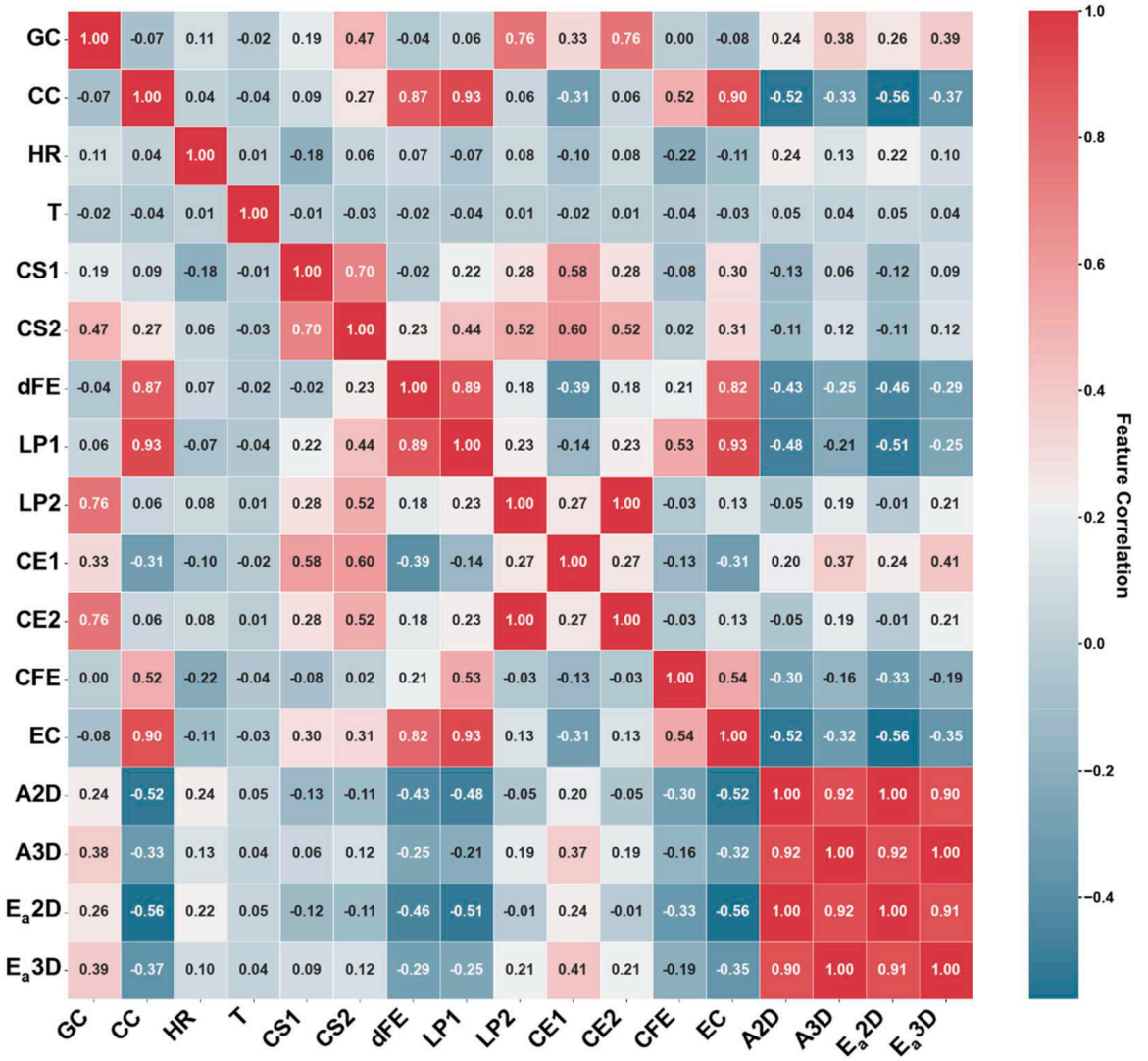


Fig. 3. Feature correlation heatmap. Feature acronyms are defined in Table 3.

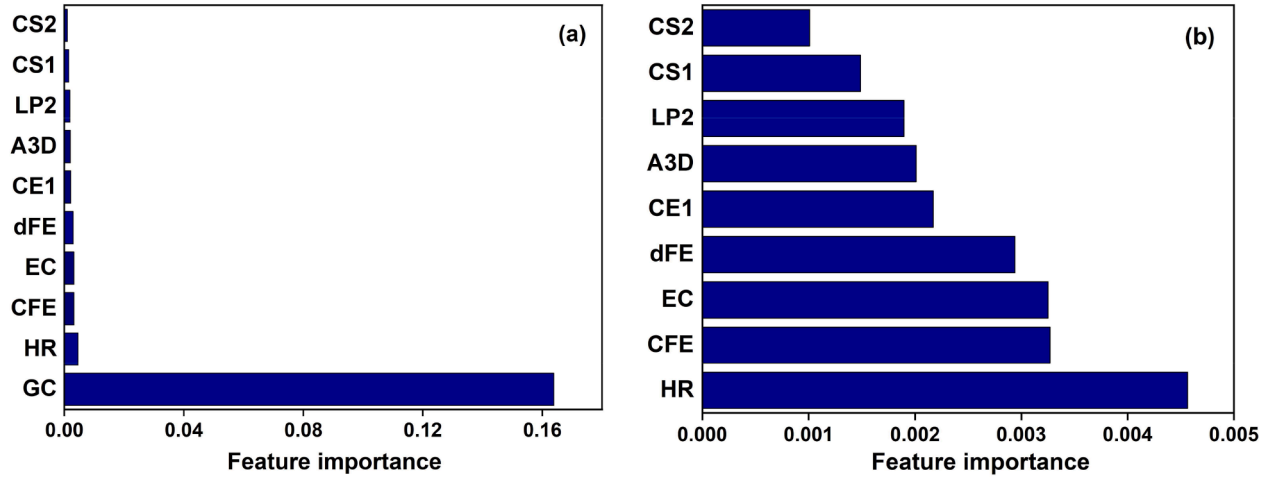


Fig. 4. (a) Random forest feature importance, excluding temperature (T) and (b) excluding temperature and Gr content (GC). Feature acronyms are defined in Table 3.

polymer during decomposition. In the PEI/Gr system, increasing GC forms a more interconnected, thermally stable lamellar network that hinders the escape of volatile products and slows oxygen diffusion into the degrading matrix. This barrier structure enhances heat localization, promotes secondary charring reactions, and stabilizes carbonaceous intermediates, ultimately increasing char yield and altering the kinetics of mass loss [57]. As a result, GC emerges as the most influential parameter in the feature importance analysis because variations in GC fundamentally reshape the decomposition pathway and exert a dominant physical control over char mass evolution across the entire temperature range. The model correctly identifies GC as a key driver due to its strong mechanistic impact on heat transfer, diffusion processes, and carbon network stabilization, all of which significantly modulate the observed thermogravimetric response.

After the exclusion of GC, heating rate (HR) becomes the most crucial factor affecting the pyrolysis of PNCs with higher heating rates known to shift the thermal decomposition peaks to higher temperatures [51]. In the presence of transition metal catalysts (Fe, Ni, and Co in this work), heating rate alters the activation of the catalyst and polymer-catalyst contact time before significant volatilization [58]. Specifically, slower heating favors polymer-catalyst interactions, possibly enhancing char formation or promoting catalytic cracking at lower temperatures, while fast heating may bypass these interactions, leading to more non-catalytic thermal decomposition pathways. At slower heating rates, Gr may stabilize the matrix via barrier effects, while at higher rates, it may enhance heat transfer, accelerating thermal decomposition [59]. Finally, at high heating rates, volatiles evolve more rapidly, possibly leading to secondary reactions such as crosslinking and carbonization in condensed phase.

Carbide formation energy (CFE) reflects the thermodynamic tendency of a transition metal catalyst to form stable metal-carbide bonds during pyrolysis, which has a direct impact on mass loss and char formation. A catalyst with lower carbide formation energy can more readily react with carbon species generated during thermal decomposition of polymer, which leads to the formation of metal carbides that stabilize the residual carbon structure. This stabilization may suppress the evolution of volatile compounds and enhance char retention, thereby mass loss is reduced [60]. In the current study, Fe exhibited the lowest carbide (Fe_2C) formation energy (Table 2), suggesting a higher affinity for bonding with carbon. This tendency was confirmed by the X-ray photoelectron spectroscopy results in our previous work [19].

The electrical conductivity (EC) of a transition metal catalyst enhances electron transfer, which can indirectly influence carbon structuring and stabilization [61]. A catalyst with higher electrical conductivity facilitates better electron mobility, which can enhance redox reactions and promote the formation of stable carbon networks during pyrolysis. This improved electron transfer capability may favor graphitization, potentially reducing the release of volatiles and, thus, lowering overall mass loss.

The number of unpaired d-orbital electrons (dFE) in transition metal catalysts plays a key role in influencing mass loss during pyrolysis by affecting the reactivity of the catalyst and its ability to interact with decomposing polymer chains [62]. Transition metal ions Fe^{3+}, Co^{2+}, and Ni^{2+} possess 5, 3, and 2 unpaired d electrons, respectively (Table 2), which determine their electron exchange capacity and catalytic behavior. In general, catalysts with higher unpaired d-electrons exhibit higher activity because of better orbital overlap and charge transfer with reactants [63], which extends to stronger interactions with polymer decomposition fragments. This behavior can promote crosslinking, suppress volatile evolution, and stabilize carbon-rich fragments, ultimately reducing the extent of mass loss and increasing char yield.

Gr and catalysts with higher cohesive energy (CE) possess more stable crystal structures, making them less likely to undergo structural changes or decomposition at elevated temperatures. For example, Gr with a CE of 7.37 (Table 2) exhibits strong atomic bonding that supports consistent catalytic interactions with the decomposing polymer matrix. In general, stable catalysts can better facilitate carbon–carbon crosslinking and suppress volatile release, leading to lower mass loss and higher char yields. In the case of kinetic parameters, only the activation energy of the 3D Avrami-Erofeev model (A3D) appears in figure importance (Fig. 4**b**), which suggests that overall, kinetic parameters are less important than the catalyst features.

Lattice parameters of Gr (LP2) generally influence its surface characteristics. Larger lattice parameters typically suggest greater interatomic distances, which can alter surface reactivity and the availability of active sites for catalytic interactions with a decomposing polymer matrix. Gr has a small LP (~ 0.25 nm in Table 2), which can promote stronger interactions with the polymer, potentially enhancing its crosslinking and stabilizing the char structure, thus reducing mass loss. Finally, the crystal system of catalyst (CS1) and Gr (CS1) (Table 2) modulates their surface properties, i.e., geometries and electronics, which determines the polymer chain decomposition mechanism, product formation, and pyrolysis efficiency. The polymer-catalyst interactions, in general, can accelerate polymer decomposition, suppress volatile loss, and enhance the formation of a stable, high-yield carbon residue. By understanding the feature importance results in this work (Fig. 4), designing a catalytic pyrolysis process to target certain char yield is facilitated. Also, by focusing on catalyst features, i.e., primarily CFE, EC, and dFE (Fig. 4**b**), a novel (or alternative) catalyst may be developed (or selected) that would balance char yield with other pyrolysis performance metrics. The developed ML model herein also supports a computational pre-screening approach before synthesis/testing.

3.2. Comparison of transformer-based neural network versus other existing models

In Fig. 5, a comprehensive comparison of the physics-informed transformer model developed herein against established baseline methods is presented. The results demonstrate that the architecture in this work achieves superior predictive performance across multiple metrics. The results in Fig. 5 were generated using five-fold cross-validation, which shows the performance of the models across five randomly assigned test-train splits.

Our model attained a mean absolute error (MAE) of 2.05±0.31 (Fig. 5**a**) and an R^2 score of 0.96±0.01 (Fig. 5**c**), substantially outperforming the standard, encoder-only transformer, LSTM, GRU, and XGBoost. Beyond absolute performance gains, a critical advantage of our physics-informed approach is the consistency of convergence, as evidenced by the substantially lower standard deviations across all metrics. While the standard transformer exhibited high variance in performance, our physics-informed variant achieved consistent results with a standard deviation of only 0.31. This stability is particularly valuable in practical applications where reliable and reproducible model behavior is essential.

A potential limitation of our ML approach is the increased training time compared to some baseline methods (Fig. 5**d**). Our model required 83.52±6.89 s per training run, notably longer than that of XGBoost (10.78±3.73 s) and moderately longer than the standard transformer (74.50±11.26 s). However, this difference should be contextualized within the broader practical considerations of model deployment. First, all training was conducted on a widely available and relatively affordable consumer-grade GPU (Nvidia RTX 4060 Laptop GPU), demonstrating that the computational requirements remain accessible and do not necessitate specialized high-performance computing infrastructure. Second, the training time remains in the order of minutes rather than hours or days, making iterative model development and hyperparameter tuning entirely feasible. Finally, when considering the overall modeling workflow, the marginal increase in training time is far outweighed by the substantial improvements in prediction accuracy and model stability. In production environments where models are typically trained once and deployed for extended periods, the one-time training cost becomes negligible relative to the ongoing benefits of superior predictive

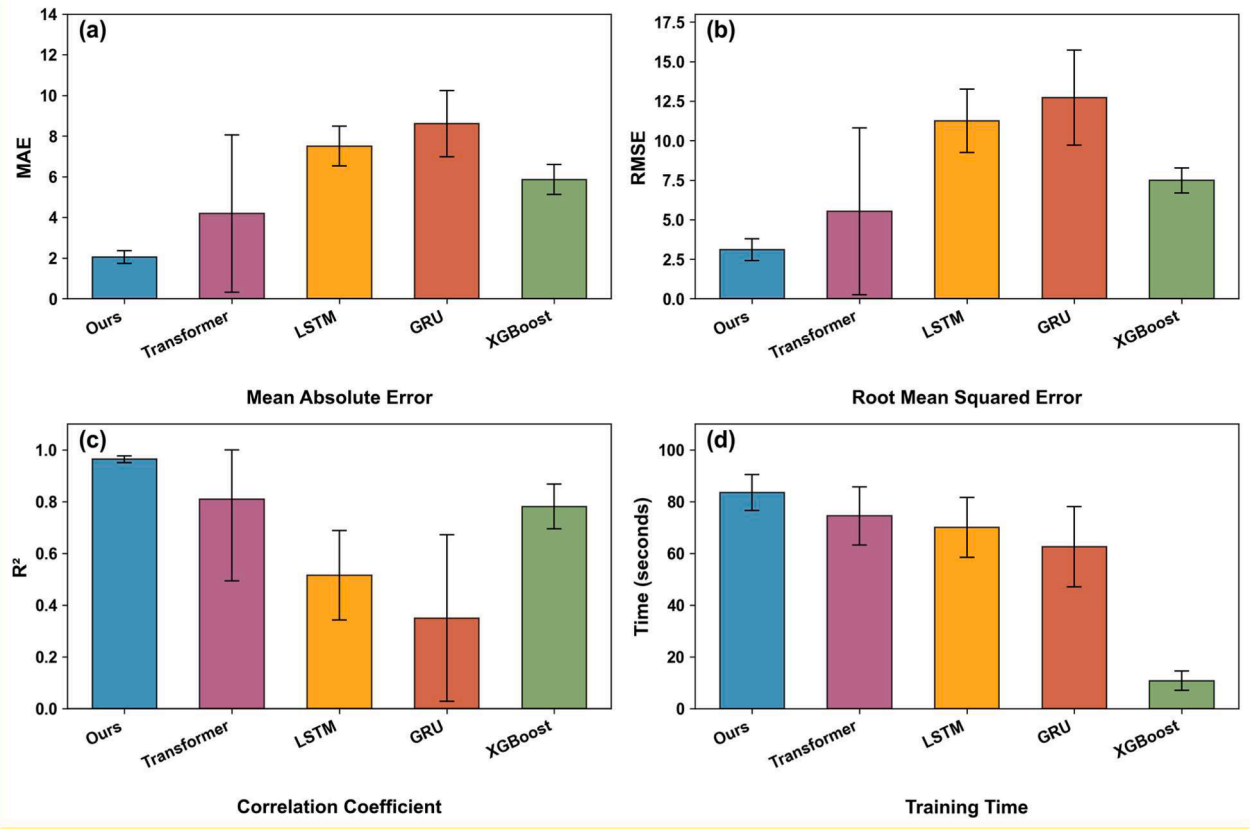


Fig. 5. Comparison of our ML architecture ("Ours"), described in Section 2.6, for char mass evolution prediction (mean ± standard deviation) with existing models across performance metrics (a) mean absolute error (MAE), (b) root mean squared error (RMSE), (c) correlation coefficient (R^2), and (d) training time.

performance. These results collectively suggest that the integration of physics-based constraints into transformer architecture provides a robust framework for achieving both high accuracy and reliable convergence, positioning this approach as a practical and effective solution for applications requiring precise and consistent predictions.

3.3. Ablation studies

To evaluate the contributions of key architectural components, four model configurations were systematically compared: (1) the full model with cross-attention and physics-informed loss (Cross-Attn + Physics), (2) cross-attention without physics-informed loss (Cross-Attn Only), (3) concatenation architecture with physics-informed loss (Concat + Physics), and (4) concatenation architecture without physics-informed loss (Concat Only). Performance metrics included MAE, root mean squared error (RMSE), R^2, and average curve-level MAE. The performance comparison across configurations is presented in Fig. 6.

The full model achieved the lowest MAE of 2.1 and RMSE of approximately 2.2, compared to 5.8 and 7.8, respectively, for the concatenation-only baseline (Figs. 6**a** and 6**b**). The full model attained $R^2 > 0.98$, substantially outperforming the concatenation-only configuration at R^2 of 0.82 (Fig. 6**c**). Notably, the error bars in Fig. 6

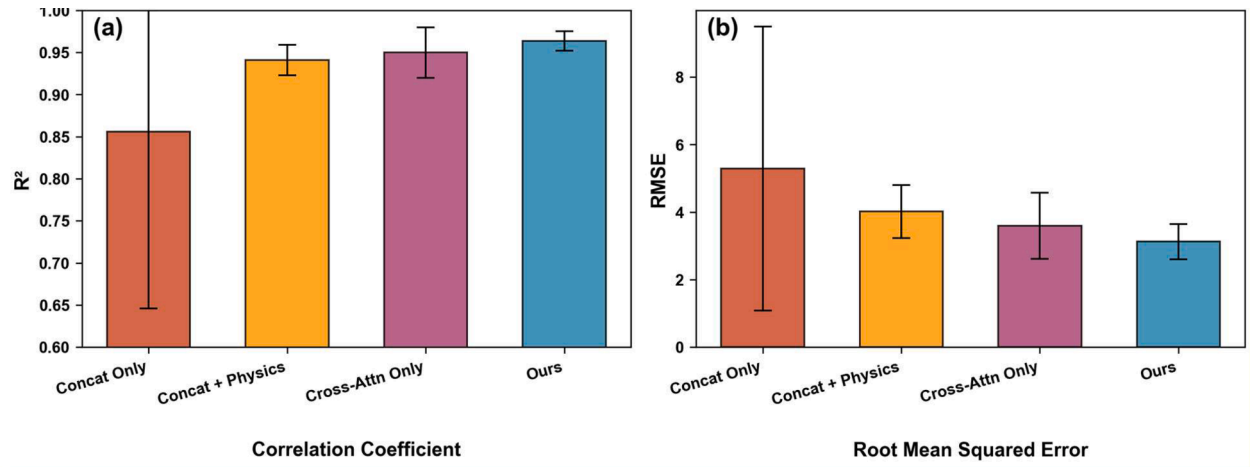


Fig. 6. Model comparisons between our model architecture ("Ours") with and without the proposed architectural components, showing model predictions (mean ± standard deviation) for encoder-only transformer networks across performance metrics (a) correlation coefficient (R^2) and (b) root mean squared error (RMSE).

demonstrate that configurations incorporating cross-attention and physics-informed loss exhibit markedly reduced variance across independent training runs, indicating improved convergence consistency. The concatenation-only baseline showed substantially larger error bars across all metrics, reflecting unstable training behavior and high sensitivity to initialization conditions.

The relative contributions of each component are quantified in Fig. 7. The cross-attention mechanism yielded an approximate 26 % improvement over concatenation (Fig. 7**a**), attributed to its ability to dynamically weight static catalyst features based on temperature-dependent decomposition stages. The physics-informed loss contributed an additional 32 % improvement (Fig. 7**b**) by enforcing mass conservation constraints, preventing non-monotonic predictions and overprediction beyond 100 % initial mass. The reduced error bars observed for models incorporating these components (Fig. 6) indicate that both the cross-attention architecture and physics-informed regularization not only improve prediction accuracy but also stabilize the optimization landscape, leading to more consistent convergence across multiple training runs. This enhanced convergence consistency is critical for reliable deployment in materials discovery workflows, where model robustness across different initializations is essential for confident predictions on unseen catalyst systems.

3.4. Prediction of the TGA curves (char evolution)

In Fig. 8, the transformer-based ML model performance with respect to the validation set is shown. The overall R^2 correlation performance of the model was 0.9836, which represents a strong prediction capability. However, the results indicate that the model predicted the thermal behavior of Sample G0-H13 with greater accuracy than that of Sample G33-H20. This discrepancy suggests that the neural network was more effective in capturing the underlying patterns and dependencies in G0-H13 (neat PEI without Gr or catalyst), potentially due to factors such as data distribution, feature representation, or sequence complexity.

The model was evaluated on a separate holdout set consisting of TGA data with a different catalyst, i.e., Co, to determine its performance on unseen catalysts. This scenario serves as a test of the ability of the model to generalize beyond its training data. The R^2 value across four TGA curves utilizing Co was 0.9776, indicating that the model was able to successfully generalize to a new catalyst. Model predictions for sample compositions containing Co and their respective R^2 values are provided in Fig. 9. The sample compositions represented the full range of Gr content (GC) (0–100 phr) and heating rates (HR) (10–17 °C/min). The high R^2 values demonstrate that the model was consistently able to accurately predict the kinetics of the pyrolysis reaction.

The predicted TGA curves follow the experimental trends across all

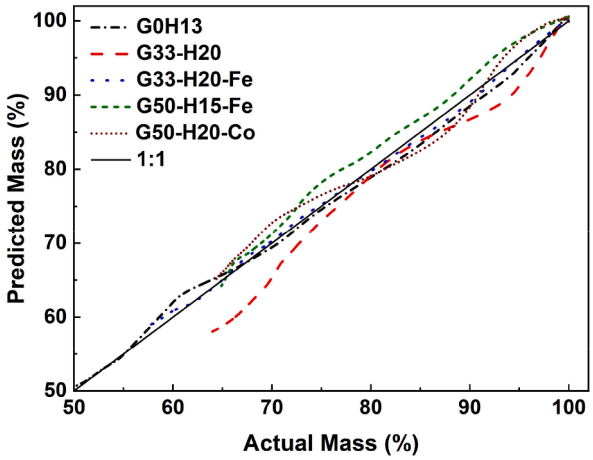


Fig. 8. Aggregated performance of the transformer-based model on the validation set (R^2 = 0.9776).

temperature ranges, capturing an onset and rate of mass loss close to the ground truth results. The char mass at high temperatures is also accurate (Fig. 9). These results are significant because the model was never trained with the data associated with the samples containing the Co catalyst, demonstrating its extrapolative capabilities.

In Fig. 10, the performance metrics of the model across each Co prediction is shown. A comprehensive understanding of the performance metrics is important to quantify sample-wise performance and failure cases.

While the model demonstrates strong overall performance on the unseen Co catalyst test set with an average R^2 of 0.9838, a detailed error analysis reveals notable variation in predictive accuracy across different sample compositions (Fig. 10). The quantitative metrics show that the model achieves the highest accuracy for Sample G100-H13-Co (R^2 = 0.991, MAE = 0.79, RMSE = 1.11), representing the formulation with maximum Gr content (100 phr) at moderate heating rate (13 °C/min). In contrast, the model exhibits reduced performance for Sample G0-H17-Co (R^2 = 0.975, MAE = 2.33, RMSE = 3.00), corresponding to neat PEI with Co catalyst at the highest heating rate (17 °C/min). This performance degradation in the absence of Gr suggests that the model relies heavily on Gr to achieve optimal predictions (as demonstrated through the feature importance in Fig. 4**a**), and that the extrapolation to high heating rates combined with minimal Gr content represents a more challenging prediction scenario. Intermediate formulations such as G33-H10-Co and G50-H20-Co exhibit moderate errors (MAE of 1.62 % and 1.43 %, respectively), suggesting that the model performs most reliably

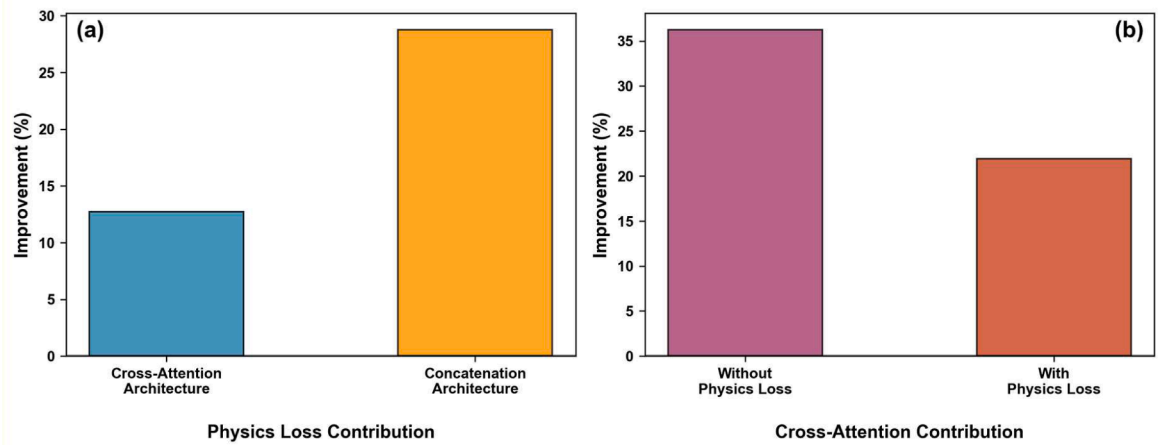


Fig. 7. Relative component contribution for (a) the physics-based loss function and (b) the cross-attention mechanism.

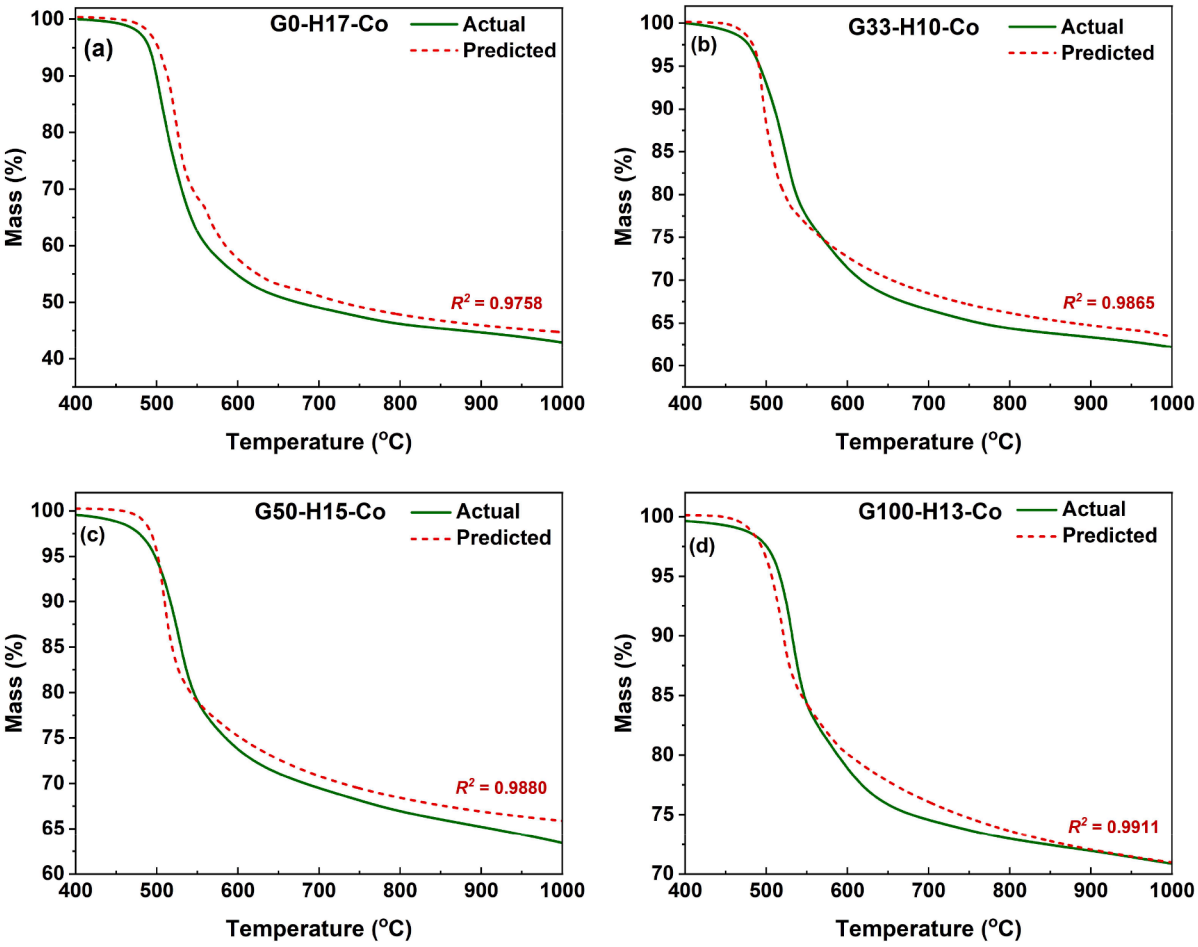


Fig. 9. Model performance in predicting the TGA curves associated (a) G0-H17-Co, (b) G33-H10-Co, (c) G50-H15-Co and (d) G100-H13-Co samples.

within the central region of the design space where Gr content and heating rate are neither at extreme values.

4. Conclusion

The study herein successfully demonstrated the capability of a physics-informed, transformer-based machine learning (ML) model to predict the char mass evolution of a polyetherimide (PEI)/graphite (Gr) nanocomposite undergoing pyrolysis in the presence or absence of a transition metal catalyst (Fe, Ni, or Co) in the temperature range of 400–1000 °C using a large TGA data set (thermograms). Apart from the main features, i.e., Gr content (GC), catalyst content (CC), heating rate (HR), and temperature (T), Gr- or catalyst-related features, i.e., crystal system (CS), d-orbital free electrons (dFE), lattice parameters (LP), cohesive energy (CE), carbide formation energy (CFE), and electrical conductivity (EC) at 20 °C, as well as 2D and 3D Avrami-Erofeev model parameters, i.e., pre-exponential factor (A2D and A3D, respectively) and activation energy (Ea_2D and Ea_3D, respectively), were used as inputs to both enhance the model and improve its extrapolative capabilities to other catalysts. A random forest-based feature importance analysis revealed GC as the most influential factor in predicting char mass evolution, with secondary contributions from HR, CFE, EC, and dFE. Model training exhibited rapid convergence, with early stopping occurring at epoch 92. The validation results showed a strong predictive performance, achieving an R^2 correlation of 0.9836, confirming the ability of the model to accurately replicate experimental TGA data. The insights drawn from feature importance analysis provides a set of principles to design catalyst systems, which may not be otherwise possible through traditional kinetic modeling alone. Further evaluation on a separate holdout set with a Co catalyst provided insight into the generalizability of the model beyond its training data. The model maintained high predictive accuracy, achieving a collective R^2 value of 0.9776 across all experiments. The best-case scenario produced near-perfect agreement with experimental results, achieving an R^2 value of 0.9911.

Our ML approach, enabling prediction of thermal decomposition for a polymer nanocomposite (PNC) precursor containing a new catalyst, provides a paradigm shift compared to traditional investigation of PNC pyrolysis. First our ML-based approach enables a low-cost alternative (few seconds on a standard GPU) to extensive experimental investigation of a TGA curve (up to several hours), dramatically accelerating the screening of a broad range of catalysts for a targeted application. Further, our ML framework in combination with optimization algorithms (e.g., Bayesian optimization or genetic algorithms) can provide a path forward for solving the inverse design problem, i.e., combinations of polymer matrix, filler content, catalyst, and processing conditions, for a desired thermal decomposition characteristic, such as maximized char yield, early degradation onset, or specific multi-step mass loss profiles. Second, unlike previously reported data-driven ML approaches, incorporation of domain knowledge through feature engineering, and introduction of a physics-informed loss function enforcing mass conservation, helps predict TGA curves with high accuracy, and improves interpretability and generalizability of the model despite the relatively low number of TGA curves in the training dataset. Third, the use of cross-attention in the transformer architecture enables dynamic weighting of static features (e.g., catalyst properties) based on temperature stage, mimicking the multiscale coupling often observed in thermochemical systems. The cross-attention mechanism allows the model to capture subtle, stage-dependent interactions between static materials features

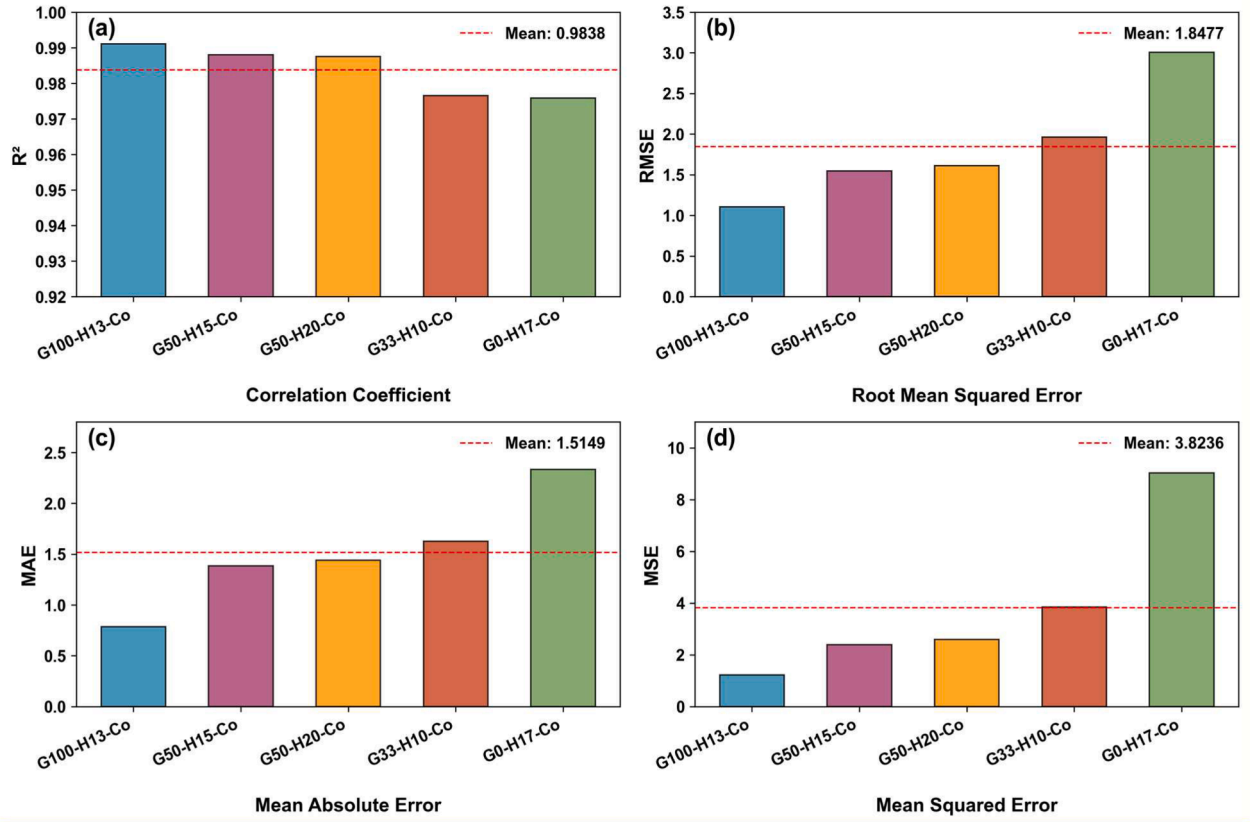


Fig. 10. Performance metrics (mean ± standard deviation) across Co hold-out experiments, including (a) correlation coefficient (R^2), (b) root mean squared error (RMSE), (c) mean absolute error (MAE), and (d) mean squared error (MSE).

and sequential thermochemical behavior. From a kinetics modeling perspective, the cross-attention mechanism is necessary because reaction rates and pathways are governed by stage-specific mechanistic regimes that are differentially influenced by material properties, requiring the model to adaptively weigh static descriptors in relation to the char mass evolution and temperature.

Despite the strength of the presented ML-guided approach to investigate pyrolysis, several limitations and open questions remain. While the ML model demonstrates excellent generalization to an unseen catalyst (Co), its predictive domain is still confined to a specific class of PNCs (PEI/Gr) and transitional metal catalysts. Generalization to other polymer matrices, filler chemistries (e.g., carbon nanotubes), or complex multi-component catalyst systems requires further work. Beyond compositional constraints, the current architecture of the model raises questions about scalability to larger, more complex systems with higher-dimensional parameter spaces, particularly when considering industrial-scale pyrolysis processes with multiple concurrent reactions and heterogeneous catalyst distributions. Additionally, the dependence of the model on accurate kinetic parameters means that predictions are only as reliable as the underlying kinetic data, which can be challenging or expensive to obtain experimentally for novel material systems. While the computational cost remains modest on consumer-grade hardware, scaling to real-time process control or high-throughput screening scenarios may require further optimization of the training pipeline and inference speed. To accomplish this, future efforts should focus on expanding the dataset to include broader ranges of polymers, alternative nanofillers, and synergistic catalyst formulations to enhance model generalizability across these systems. Moreover, incorporating additional features such as molecular descriptors, catalyst surface properties, and reactor geometry parameters may improve predictive accuracy and extend applicability. Particularly promising directions include the development of transfer learning strategies to leverage knowledge from well-characterized systems when predicting behavior of novel materials, integration of real-time temperature and gas composition data for adaptive process control, and the incorporation of uncertainty quantification methods to provide confidence intervals for predictions in extrapolative regimes. Finally, coupling the ML framework with mechanistic reaction network models could enable interpretation of catalyst effects at a molecular level while maintaining the computational efficiency advantages of data-driven approaches.

CRediT authorship contribution statement

Masoud Salavati: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Stanford White:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mohammed Majdoub:** Writing – review & editing, Data curation. **Mine G. Ucak-Astarlioglu:** Writing – review & editing, Project administration. **Ahmed Al-Ostaz:** Writing – review & editing, Resources, Funding acquisition. **Samrat Choudhury:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization. **Sasan Nouranian:** Writing – review & editing, Validation, Supervision, Resources, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Sasan Nouranian reports financial support was provided by US Army Engineer Research and Development Center. Samrat Choudhury reports financial support was provided by National Science Foundation. Stanford White IV reports financial support was provided by National Science Foundation. If there are other authors, they declare that they have

no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.chphi.2025.100994.

Data availability

Data will be made available upon request.

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